

Initiation à R : prise en main

- ✓ Découvrir **R & R Studio** et savoir comment les installer

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Pourquoi utiliser R ?

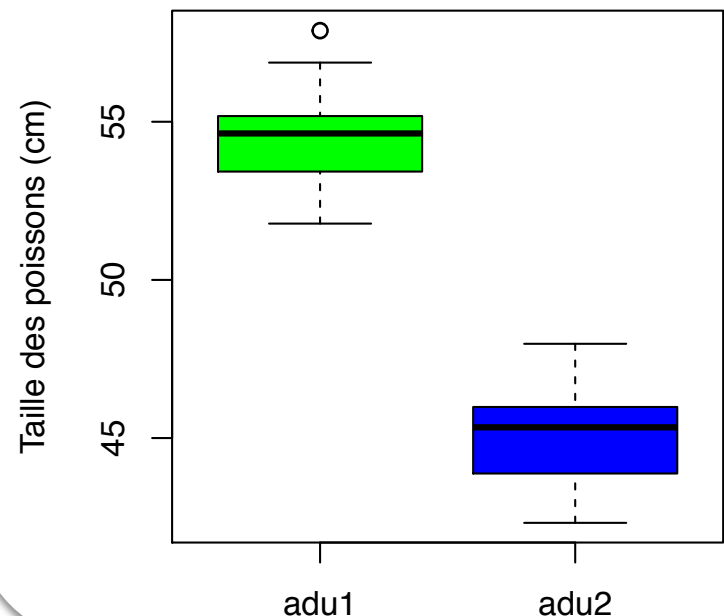
Conduire des analyses sur vos données et les visualiser

```
> t.test(a, b, var.equal=TRUE)
```

Two Sample t-test

data: a and b

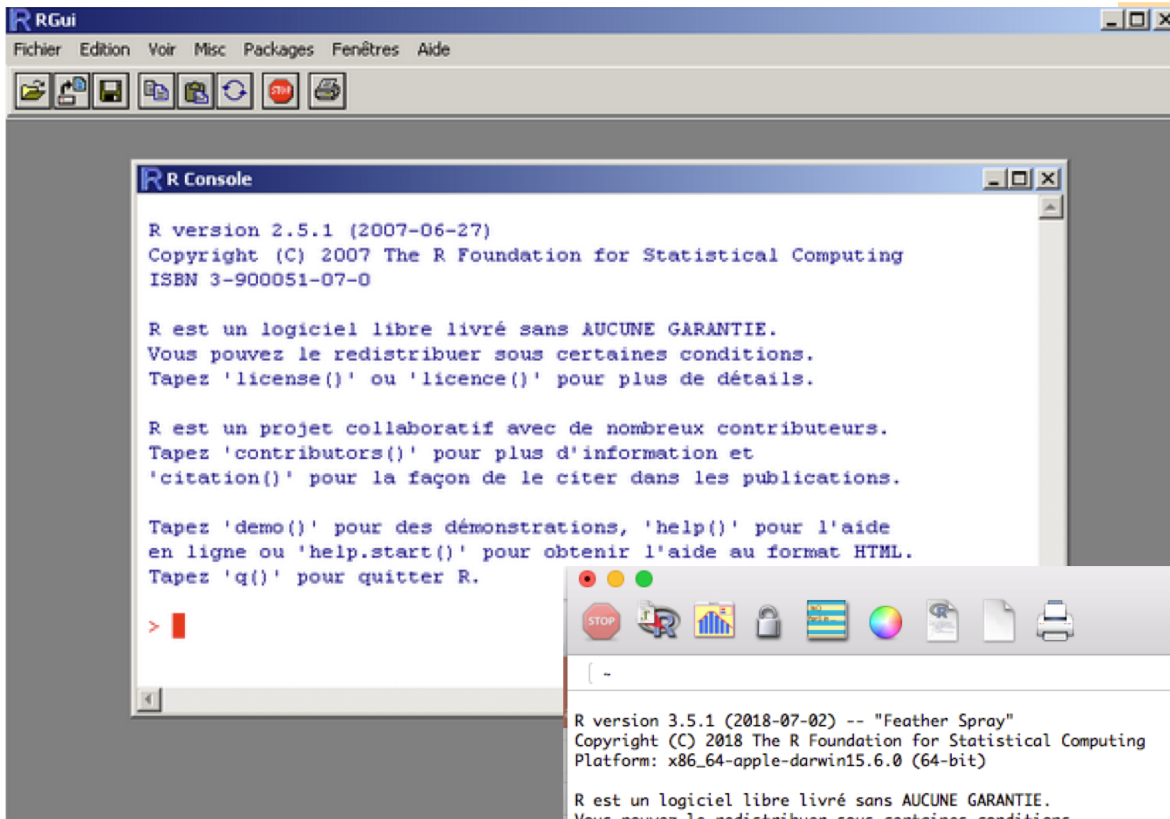
t = 2.7484, df = 48, **p-value = 0.008414**
alternative hypothesis: true difference in mean



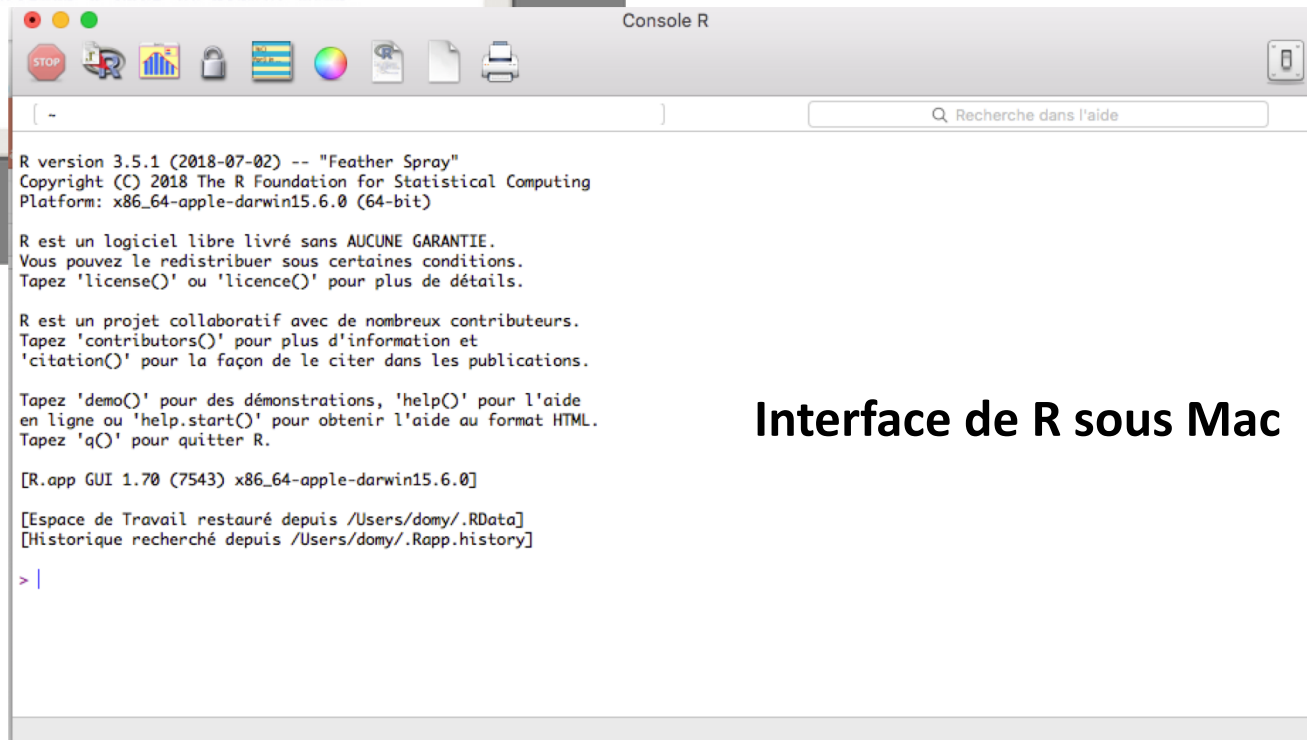


Pourquoi utiliser R ?

- R est un **langage de programmation** développé pour l'analyse statistique
- disponible **pour toutes les plateformes**   
- **libre & gratuit**
- **puissant et flexible** : les fonctionnalités de base peuvent être étendues à l'aide d'extensions, appelées **packages** (ou librairies)
- en **constante évolution**
- **communauté d'utilisateurs importante** : la plupart de vos questions trouveront une réponse : vidéos, tutoriels, sites internet dédiés, forums...



Interface de R sous Windows



Interface de R sous Mac



Pourquoi utiliser R Studio ?

- Environnement de développement intégré (IDE) de R
- Affichage simultané des différentes fenêtres : éditeur de script + console + fichiers graphiques + page d'aide +



Interface de R Studio



Installer R

<http://cran.r-project.org/>

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2013-09-25, Frisbee Sailing) [R-3.0.2.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha and beta releases](#) (daily snapshots, created only in time periods before a planned release).

Choisir l'installateur correspondant à votre système d'exploitation



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Installer RStudio

<http://www.rstudio.com>



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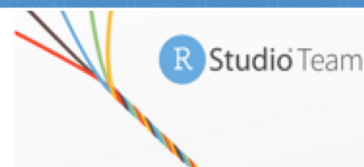
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Download the RStudio IDE

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The RStudio IDE is a set of integrated tools designed to help you be more productive with R and Python. It includes a console, syntax-highlighting editor that supports direct code execution, and a variety of robust tools for plotting, viewing history, debugging and managing your workspace.

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<https://rstudio.cloud/>



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